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Dual UAV-Assisted Secure IoT Networks: Resource Allocation and 3D Trajectory Design

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***Abstract*—This paper investigates a dual unmanned aerial vehicle (UAV)-assisted communication system for enabling secure information transfer between a base station (BS) and an Internet of Things device against randomly located terrestrial eavesdroppers, modeled as a homogeneous Poisson point process. In this setup, both the UAV relay and the jammer harvest energy from the radio frequency signals transmitted by the BS and utilize it for relaying and transmitting jamming signals. We formulate the average secrecy rate maximization problem by considering the time-splitting ratio, transmit powers, and the three-dimensional trajectories of UAVs. Moreover, we account for practical constraints such as energy harvesting and information causality, UAVs' collision avoidance, limited battery capacity, etc. Considering the system dynamics, we reformulate the problem as a constrained Markov decision process and solve it with a soft actor-critic (SAC) algorithm. Specifically, the SAC agent learns an optimal policy by observing time-varying channel gains and spatial randomness of homogeneous Poisson point process eavesdroppers. Lastly, we demonstrate that our framework significantly enhances ASR performance compared to benchmark schemes through numerical analysis.**

***Index Terms*—UAVs, cooperative jamming, HPPP eavesdroppers, 3D trajectory design, and deep reinforcement learning.**

I. Introduction

Wireless technology is advancing rapidly, with 6G research uncovering new possibilities across land, air, and sea to ensure extensive access and reliable performance [1], [2]. Integrating unmanned aerial vehicles (UAVs) with ultradense and heterogeneous Internet of Things (IoT) devices offers a promising solution for many applications [3]. However, a significant concern for UAV communications is their broadcast nature, which makes them vulnerable to eavesdropping by malicious entities [4]. In this vein, physical layer security (PHY security) has been widely employed to secure communications by leveraging wireless channels' random nature [5]–[8]. For instance, block coordinate descent (BCD) and successive convex approximation (SCA) methods are used in [6], [7] to solve the non-convex secrecy rate maximization problem for dual UAV-assisted networks. Another challenge arises from the energy limitations of battery-powered UAVs, which restrict their operational lifespan. UAVs that function as aerial relays or jammers need extra energy to operate effectively. However, onboard batteries' limited size, weight, and power constrain the overall operational time [9]. Radio-frequency (RF) energy harvesting (EH) techniques have potential to extend the lifespan of wireless networks.

In this direction, [10] and [11] considered RF energy harvesting at energy-constrained terrestrial nodes and UAVs, respectively, aiming to maximize secrecy energy efficiency. However, the works in [6]–[8], [10], [11] relied on traditional optimization techniques, often falling short in dynamic environments. In contrast, deep reinforcement learning (DRL) has gained attention for its model-free decision-making capabilities [1], [12]. For example, the work in [13] aimed to enhance secrecy rates for mobile ground nodes by framing the problem as a Markov decision process (MDP) and addressing it with the DRL algorithm. However, [6]–[8], [10], [13] focused solely on security issues for the air-to-ground (A2G) link.

Motivated by the existing literature gaps, we utilize a dual UAV-assisted relaying and cooperative jamming scheme for secure communications between the base station (BS) and IoT device. In the proposed framework, the relay and jammer UAVs harvest energy from RF signals transmitted by the BS. The UAV relay forwards the information signal to the IoT device, while the jammer UAV transmits jamming signals to degrade the eavesdroppers' wiretap links. The major contributions of this paper are as follows:

- We consider spatially random homogeneous Poisson point process (HPPP) eavesdroppers that can intercept A2G and ground-to-air (G2A) links.
- We maximize average secrecy rate (ASR) by jointly optimizing the time-switching ratio (TSR), transmit powers and 3D trajectories of UAV under various practical constraints.
- To handle the non-convexity and stochasticity of the optimization problem, we reformulate it as a constrained MDP and solve using soft actor-critic (SAC) algorithm.
- We show our framework's effectiveness by comparing it with four benchmark schemes in simulation results.

For clarity, we provide Table I to highlight the contributions made in this work and compare them with relevant studies.

II. System Model and Problem Formulation

A. System Model Discussion

As depicted in Fig. 1, we consider a dual-UAV-assisted secure network where a BS (B) communicates with an IoT

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TABLE I
COMPARISON OF PROPOSED WORK WITH RELEVANT LITERATURE WORKS.

Works	System Model					Causality Constraints	Collision Avoidance	EH Parameter Optimization	Solution Approach
	Aerial Links	Number of Eavesdroppers	Channel Model	EH	Eavesdroppers' Uncertain Location				
[7]	A2G	Single	FSPL	×	✓	Information	✓	×	SCA, BCD
[10]	A2G	Two	Rician	✓	✓	×	×	✓	SCA, BCD
[11]	A2G, G2A	Multiple	Rician	✓	✓	Energy	×	✓	SCA, BCD
[13]	A2G/G2A	Multiple	FSPL	×	×	×	×	×	DDPG
This work	A2G, G2A	Multiple	FSPL, Rician, Rayleigh	✓	HPPP	Both	✓	✓	CSAC
Free-space path loss (FSPL)			Covered (✓)				Not covered (×)		

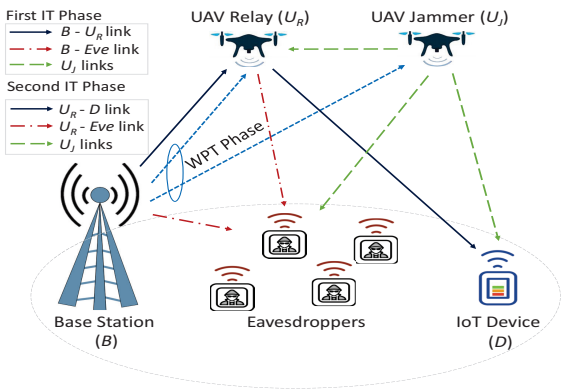

Fig. 1. Dual UAV-enabled secure IoD communications.

device (D) via UAV relay (U_R) that assists in information transfer (IT) from B to D using decode-and-forward relaying. Multiple terrestrial eavesdroppers, modeled as an HPPP denoted as Φ_e with the density of λ_e, are located within a disc of radius R_e and can wiretap the information. Herein, the kth eavesdropper is denoted as E_k, $\forall E_k \in \Phi_e$ [14]. Therefore, a UAV jammer (U_J) broadcasts jamming signals to degrade the eavesdroppers' wiretap links [7]. We employ trajectory discretization for planning UAVs' path by dividing the total flight duration T into N time slots with equal intervals of $\Delta = T/N$. Importantly, Δ is chosen small enough to make the channel state information of these nodes static within each time slot $[n]$, $\forall n \in \{1, 2, \ldots, N\}$.

A transmission block is divided into three phases, i.e., a wireless power transfer (WPT) of $\alpha[n]\Delta$ duration and two IT phases of $\frac{(1-\alpha[n])\Delta}{2}$ duration, where $\alpha[n] \in [0, 1]$ is the TSR. In the WPT phase, B transmits the RF signal to both UAVs that harvest the energy and utilize it solely for relaying and transmitting jamming signals [11]. Then, B transmits the information signal to U_R in the first IT phase, and U_R relays the signal to D in the second IT phase. The legitimate nodes' locations are given by two-dimensional (2D) Cartesian coordinates, i.e., $w_t = (x_t, y_t)$, $\forall t \in \{B, D\}$. The 3D time-varying positions of U_ϱ are denoted as $Q_\varrho[n] = (q_\varrho[n], H_\varrho[n])$, where $q_\varrho[n] = (x_\varrho[n], y_\varrho[n])$ is the horizontal coordinates and $H_\varrho[n]$ denotes the flying altitude of UAVs for $\varrho \in \{R, J\}$. UAVs can adjust their position in between two successive time slots with maximum flight velocity denoted as $v_{\varrho,max}$ [8]. The aerodynamic power consumption of UAVs in the nth slot is expressed as $P_{\varrho,f}[n] = \left(P_0\left(1+\frac{3\nu_\varrho^2[n]}{\mathcal{S}_{\text{tip}}^2}\right)+P_i v_0 \nu_\varrho[n]+\frac{1}{2}d_0\mu s A \nu_\varrho^3[n]\right)$, where P_0, P_i, $\mathcal{S}_{\text{tip}}$, v_0 d_0, s, μ, and A are the propulsion parameters of UAVs. It is important to note that the onboard UAVs' batteries power propulsion and flight control [6], [15].

The time-varying channel gains for $B-U_\varrho$, $U_\varrho - D$ and $U_\varrho - E_k$ links are given as $g_{B\varrho}[n] = h_{B\varrho}[n]|\tilde{h}_{B\varrho}[n]|^2$, $g_{\varrho l}[n] = h_{\varrho l}[n]|\tilde{h}_{\varrho l}[n]|^2, \forall\ l \in \{D, E_k\}$, respectively, where $h_{B\varrho}[n] = \beta_0 d_{B\varrho}^{-2}[n]$ and $h_{\varrho l}[n] = \beta_0 d_{\varrho l}^{-2}[n]$ are the large scale fading coefficients. The distance between corresponding nodes are given by $d_{B\varrho}[n] = (\|w_B - q_\varrho[n]\|^2 + H_\varrho^2[n])^{\frac{1}{2}}$ and $d_{\varrho l}[n] = (\|q_\varrho[n] - w_l\|^2 + H_\varrho^2[n])^{\frac{1}{2}}$ and β_0 denotes the reference channel power gain at $1\,\text{m}$ unit distance. Whereas, the small scale fading coefficients are given as $\tilde{h}_{B\varrho}[n] = \sqrt{\frac{K_u[n]}{K_u[n]+1}} + \sqrt{\frac{1}{K_u[n]+1}}\check{h}_{B\varrho}$ and $\tilde{h}_{\varrho l}[n] = \sqrt{\frac{K_u[n]}{K_u[n]+1}} + \sqrt{\frac{1}{K_u[n]+1}}\check{h}_{\varrho l}$, where $K_u[n]$ denotes the Rician factor, and $\check{h}_{B\varrho} = \check{h}_{\varrho l} \sim \mathcal{CN}(0, 1)$ are complex Gaussian random variable [11]. Furthermore, we consider the Rayleigh fading scenario for the terrestrial communication links and define channel power gain as $g_{BE_k} = \beta_0 d_{BE_k}^{-\gamma_0}\psi$ for $B-E_k$, where $d_B = \|w_B - w_{E_k}\|$ represents the distance between these nodes, $\gamma_0 \geq 2$ is path loss exponent and ψ is exponentially distributed random variable [15].

B. Wireless Power and Data Transmission

In the WPT phase, energy harvested by the UAVs in $[n]$th time slot is given by $E_\varrho[n] = \eta P_B g_{B\varrho}[n]\alpha[n]\Delta$, where P_B is the transmit power of B, η denotes EH circuitry's energy conversion efficiency. Consequently, powers available at UAVs are $P_R[n] = \frac{2\eta P_B \alpha[n]\Delta g_{BR}[n]}{1-\alpha[n]}$ and $P_J[n] = \frac{\eta P_B \alpha[n]\Delta g_{BJ}[n]}{1-\alpha[n]}$, respectively. In the first IT phase, B transmits the information signal to U_R, and U_J transmits a jamming signal to degrade the wiretap link of E_k. The signals received by U_R and E_k are given as $z_{Bo}[n] = \sqrt{P_B g_{Bo}[n]}u_B[n] + \sqrt{P_J[n] g_{Jo}[n]}u_J[n] + n_o[n]$, $\forall o \in \{R, E_k\}$, where u_B and u_J are unit energy information and jamming signals of B and U_J, respectively and $n_o \sim \mathcal{CN}(0, N_o)$ denote the additive white Gaussian noise (AWGN) at U_R and E_k. In the second phase of IT, U_R decodes the information received from B and forwards it to D. However, E_K can wiretap by exploiting the A2G link of U_R. U_J transmits jamming signals to E_K to degrade the wiretap link. Therefore, the signal received by D and E_k from U_R are given as $z_{Rl}[n] = \sqrt{P_R[n] g_{Rl}[n]}u_R[n] + \sqrt{P_J[n] g_{Jl}[n]}u_J[n] + n_l[n]$, where the transmit power and unit energy symbol of U_R are denoted as $P_R[n]$ and z_R, respectively, and $n_l \sim \mathcal{CN}(0, N_l)$ is the AWGN at D and E_k.

Notably, U_R and D can eliminate the jamming signal from the received signals $z_{Bo}[n] and z_{Rl}[n]$, whereas E_k is

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unable to eliminate the jamming signal while extracting the information [11], [16]. Due to small-scale fading coefficients, the achievable rates are random and typically involve expectations over channel realizations. Therefore, $B-D$ link's achievable data rate is given by $R_l[n]=\mathbb{E}\big[\frac{(1-\alpha[n])\Delta}{2}\log_2(1+\min(\gamma_{BR}[n],\gamma_{RD}[n]))\big]$. We consider information causality in the system, which states that $\sum_{n=1}^{m-1}R_{BR}[n]\geq\sum_{n=2}^{m}R_{RD}[n],\ \forall m\in\{2,\cdots N\}$, where $\gamma_{BR}[n]=\frac{P_B g_{BR}[n]}{N_R}$ and $\gamma_{RD}[n]=\frac{P_R[n]g_{RD}[n]}{N_D}$ are the signal-to-noise ratio (SNR) and $R_{BR}[n]=\mathbb{E}\big[\frac{(1-\alpha[n])\Delta}{2}\log_2(1+\gamma_{BR}[n])\big]$ and $R_{RD}[n]=\mathbb{E}\big[\frac{(1-\alpha[n])\Delta}{2}\log_2(1+\gamma_{RD}[n])\big]$ denote the achievable rate from $B-U_R$ and U_R-D in time slot $[n]$. We focus on the worst-case scenario, where a colluding eavesdroppers can intercept the information from both IT phases. Therefore, the eavesdropping rate is given by $R_w[n]=\mathbb{E}[\frac{(1-\alpha[n])\Delta}{2}\log_2(1+\sum_{E_k\in\Phi_e}(\gamma_{BE_k}[n]+\gamma_{REk}[n]))]$, where $\gamma_{BE_k}=\frac{P_B g_{BE_k}}{P_J g_{JE_k}[n]+N_{E_k}}$ and $\gamma_{RE_k}[n]=\frac{P_R[n]g_{RE_k}[n]}{P_J[n]g_{JE_k}[n]+N_{E_k}}$ are the SNR for $B-E_k$ and U_R-E_k, respectively. Thus, the secrecy rate can be expressed as

$$R_s[n]=[R_l[n]-R_w[n]]^+,\quad \text{where } [f]^+\triangleq\max\{f,0\}. \tag{1}$$

C. ASR Maximization Problem

In this section, we formulate the ASR maximization problem (P1) by jointly optimizing TSR (α), transmit powers ($P_\varrho[n]$), and 3D trajectories of U_R and U_J ($Q_\varrho[n]$). The ASR maximization problem is defined as

$$\text{(P1)}:\ \max_{\alpha[n],P_\varrho,Q_\varrho[n]}\ \frac{1}{N}\sum_{n=1}^{N}R_s[n] \tag{2a}$$

$$\textbf{st.}\quad 0\leq\alpha[n]\leq 1,\ \forall n, \tag{2b}$$

$$\sum_{n=1}^{N}E_\varrho[n]\geq\sum_{n=1}^{N}(P_\varrho[n]\kappa_\varrho(1-\alpha[n])\Delta),\ \forall n, \tag{2c}$$

$$0\leq P_\varrho^{\text{avg}}\leq\beta_\varrho^{\max},\ \forall n, \tag{2d}$$

$$\sum_{n=1}^{m-1}R_{BR}[n]\geq\sum_{n=2}^{m}R_{RD}[n],\ \forall m,n, \tag{2e}$$

$$q_{\varrho_0}=q_\varrho[0],\ q_{\varrho_F}=q_\varrho[N], \tag{2f}$$

$$H_{\varrho,\min}[n]\leq H_\varrho[n]\leq H_{\varrho,\max}[n], \tag{2g}$$

$$H_\varrho[n+1]-H_\varrho[n]\leq v_{\varrho_{\text{move}}}, \tag{2h}$$

$$q_\varrho[n+1]=q_\varrho[n]+v_{\varrho,\max}[n]\Delta,\ \forall n, \tag{2i}$$

$$x_\varrho^2[n]+y_\varrho^2[n]\leq r_{\varrho,\max}^2,\ \forall n, \tag{2j}$$

$$\|q_R[n]-q_J[n]\|^2\geq d_{\min},\ \forall n, \tag{2k}$$

$$\|q_{\varrho_F}-q_{\varrho_1}\|\leq v_{\varrho,\max}[n]T,\quad\forall n, \tag{2l}$$

$$R_l[n]-R_w[n]>0,\quad\forall n, \tag{2m}$$

where $\kappa_\varrho=1/2$ for $\varrho=R$ and $\kappa_\varrho=1$ for $\varrho=J$. We incorporate various practical constraints, including TSR, energy and information causality, UAV power budgets, and limitations on UAV trajectories and operations. Specifically, (2b) imposes the TSR constraint, while the UAVs' energy causality constraints are defined in (2c). Moreover, the average power budgets for the UAVs are given in (2d), denoted by $\beta_\varrho^{\max}$, which must be greater than their average power consumption, expressed as $P_\varrho^{\text{avg}}=\frac{1}{N}\sum_{n=1}^{N}\big(P_\varrho[n]+P_{\varrho,f}[n]\big)$. The information causality constraint at U_R is given by (2e) [8]. Next, the restrictions on UAVs' initial and final locations are given by (2f). Then, (2g) and (2h) impose the constraints on UAVs' vertical motions. The constraint on UAVs' flight velocity is defined in (2i). Further, the constraint on UAVs' maximum flight radius is given in (2j), where $r_{\varrho,\max}$ denotes the maximum flight radius of UAVs. Collision avoidance requires a minimum distance ($d_{\min}$) between UAVs, which is imposed by (2k). Then, (2l) ensures that UAVs reach their final positions within time T. The operator $[\cdot]^+$ present in (1) makes it a non-smooth function, and therefore, we impose constraint (2m) to guarantee non-negative ASR values.

It is challenging to solve (P1), as the objective function in (2a) exhibits non-convexity due to the tightly coupled variables, such as $\alpha[n]$ and $(Q_\varrho[n])$. Also, the presence of the expectation operator in the rate expressions of (2a), (2e), and (2m) makes (P1) intractable. The inclusion of HPPP eavesdroppers introduces a semi-infinite number of constraints in (2a) and (2m), further complicating the optimization problem's tractability. To address the system dynamics and stochastic elements, we reformulate the ASR maximization problem as a CMDP and solve it using the SAC algorithm.

III. DRL-APPROACH FOR ASR MAXIMIZATION PROBLEM

We transform the original ASR maximizing problem into a CMDP, which constitutes of the following elements $\big(\mathcal{S},\mathcal{A},P,R,\{C_k\}_{k=1}^{7},\gamma,d\big)$ [17], which are described as

1) State Space (S): The system's state space in the nth time slot, denoted by $s[n]$, includes all time-varying parameters relevant to decision-making. Specifically, $s[n]=[Q_\varrho[n],P_\varrho[n],\alpha[n],\nu_\varrho[n],e_\varrho[n],\tau_\varrho[n],g_{ij}[n]]$, where $e_\varrho[n]=e_\varrho[n-1]+\eta P_B g_{B\varrho}[n-1]\alpha[n-1]\Delta-(\frac{(1-\alpha[n-1])\Delta}{2}P_\varrho[n-1]+P_{\varrho,f}[n-1]\Delta)$, $n\geq 1$ with $e_\varrho[0]=\beta_\varrho^{\max}$ represent the residual energy in the UAVs' batteries. The remaining travel time slack to reach their respective final positions within the episode duration is captured by $\tau_\varrho[n]=(N-n)-\frac{\|q_{\varrho_F}-q_\varrho[n]\|}{v_{\varrho,\max}\Delta}$, where q_{ϱ_F} denotes the target position of U_ϱ and $v_{\varrho,\max}$ is its maximum speed. The state also implicitly reflects the influence of channel conditions and environmental unpredictability. The wireless channel gains $g_{ij}[n]\in\{g_{BR}[n],g_{BJ}[n],g_{RD}[n],g_{BE_k}[n],g_{RE_k}[n]\}$ are subject to both large-scale path loss and small-scale fading, also implicitly capturing the spatial randomness and varying channel conditions of the HPPP-distributed eavesdroppers.

2) Action Space (A): The continuous actions taken by the agent in the nth slot comprises of $a[n]=[\Delta q_\varrho[n],\Delta H_\varrho[n],P_\varrho[n],\alpha[n]]\in\mathcal{A}$, where $\Delta q_\varrho[n]\triangleq q_\varrho[n+1]-q_\varrho[n]\in\mathbb{R}^2$ and $\Delta H_\varrho[n]\triangleq H_\varrho[n+1]-H_\varrho[n]\in\mathbb{R}$ represent the commanded horizontal and vertical displacements for UAVs. The mobility commands must satisfy the kinematic constraints given as $\|\Delta q_\varrho[n]\|\leq v_{\varrho,\max}\Delta$ and $|\Delta H_\varrho[n]|\leq v_{\varrho,\max}\Delta$. Executing $a[n]$ deterministically updates the UAV positions through $q_\varrho[n+1]=q_\varrho[n]+\Delta q_\varrho[n]$ and $H_\varrho[n+1]=H_\varrho[n]+\Delta H_\varrho[n]$. Moreover, the agent jointly selects $\alpha[n]$ with the trajectory motion under the instantaneous

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residual energies $e_\varrho[n]$ and the average-power constraint. The small-scale fading and the spatial randomness of the HPPP eavesdroppers are determined by the environment's stochastic transition and reward functions.

3) State Transition Probability (P): The transition probability $P(s[n+1]|s[n],a[n])$ is influenced by UAVs' mobility constraints and stochastic wireless channels, which are affected by small-scale fading and the random spatial distribution of eavesdroppers Φ_e. The displacements, $\Delta q_\varrho[n]$ and $\Delta H_\varrho[n]$, update the UAVs positions within specified velocity limits. Meanwhile, the environment generates small-scale fading coefficients $\tilde{h}_{B\varrho}[n]$, $\tilde{h}_{\varrho l}[n]$ in each time-slot and uses the λ_e in each episodes to compute $g_{B\varrho}[n+1]$ and $g_{\varrho l}[n+1]$.

4) Reward Function (R): The reward function, designed to steer the agent towards energy–aware, time-efficient, and ASR-maximizing behavior, is expressed as

$$R[n]=\begin{cases}\lambda_1 e_\varrho[n]+\lambda_2\tau_\varrho[n]+\lambda_3 R_s[n], & \tau_{\min}[n]<0,\\ \lambda_1 e_\varrho[n]+\lambda_2\tau_\varrho[n]+\lambda_3 R_s[n]+\lambda_4, & \text{otherwise},\end{cases} \tag{3}$$

where λ_1, λ_2, λ_3, and λ_4 are the positive weights and a terminal bonus λ_4 is awarded when both UAVs can still reach their targets within the remaining time.

5) Cost signals: In addition to the reward $R[n]$, the environment returns the cost vector in the nth slot, which is described as $C_1[n]=\frac{(1-\alpha[n])\Delta}{2}P_R[n]+P_{R,f}[n]\Delta$, $C_2[n]=\frac{(1-\alpha[n])\Delta}{2}P_J[n]+P_{J,f}[n]\Delta$, $C_3[n]=P_R[n]+P_{R,f}[n]$, $C_4[n]=P_J[n]+P_{J,f}[n]$, $C_5[n]=\left[\sum_{t=1}^{m-1}R_{BR}[t]-\sum_{t=2}^{m}R_{RD}[t]\right]^+$, $C_6[n]=\left[\|q_R[n]-q_J[n]\|^2-d_{\min}^2\right]^+$, $C_7[n]=\left[R_l[n]-R_w[n]\right]^+$. The averaged costs must satisfy $\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}^\pi[C_\kappa[n]]\leq d_\kappa,\forall\kappa=\{1,\dots,7\}$, where $d_1=\sum_{n=1}^{N}E_R[n]$, $d_2=\sum_{n=1}^{N}E_J[n]$, $d_3=\beta_R^{\max}$, $d_4=\beta_J^{\max}$, and $d_5=d_6=d_7=0$.

6) Policy (π): Control policy $\pi:\mathcal{S}\rightarrow\mathcal{A}$ defines a mapping from states to actions, determining the UAVs' trajectory, power, and TSR at every time slot. The objective is to find the optimal policy π^* that maximizes the expected cumulative discounted reward over the entire flight duration T is $\pi^*=\arg\max_\pi\mathbb{E}^\pi\Big[\sum_{n=1}^{N}\gamma^{n-1}R[n]\Big]$, s.t. $\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}^\pi[C_\kappa[n]]\leq d_\kappa$, where $\gamma\in(0,1]$ is the discount factor.

A. Proposed CSAC-based Framework

To solve the CMDP mentioned above, we utilize the CSAC algorithm, which is obtained by merging the standard SAC algorithm with the Lagrangian relaxation framework. It is an off-policy, entropy-regularized actor-critic algorithm that augments classical learning method and policy-entropy term [17]. It has fewer sensitive hyperparameters and provides stability than DDPG [13] and DQN [18]. The standard SAC algorithm maximizes $\mathcal{J}(\pi_\xi)=\mathbb{E}_{\pi_\xi}\Big[\sum_{n=0}^{N-1}\gamma^n\big(R(s[n],a[n])+\alpha\mathcal{H}(\pi_\xi(\cdot|s[n]))\big)\Big]$, where $\pi_\xi(a|s)$ denotes a stochastic policy having neural network with weights ξ, $\alpha>0$ is the temperature factor that balances exploitation and exploration, $\mathcal{H}(\pi)=-\mathbb{E}_{a\sim\pi}[\log\pi(a|s)]$ is the policy's Shannon entropy. However, the policy must also satisfy seven hard constraints for the considered CMDP. Simple action clipping cannot enforce these requirements because they involve averaged components and a couple of states across multiple slots. The CSAC algorithm takes instantaneous cost and dual multiplier vectors, defined as $\mathbf{C}=[C_1,\dots,C_7]^\mathsf{T}$, and $\lambda=[\lambda_1,\dots,\lambda_7]^\mathsf{T}\geq 0$, respectively. By combining the entropy regularizer and reformed CMDP objective, it becomes unconstrained maximization problem by minimizing with respect to λ, given as

$$\min_\lambda\ \max_{\pi_\xi}\mathbb{E}_{\pi_\xi}\Big\{\sum_{n=0}^{N-1}\gamma^n\Big(\big(R(s[n],a[n])-\sum_{\kappa=1}^{7}\lambda_\kappa C_\kappa\big)+\alpha\mathcal{H}(\pi_\xi(\cdot|s[n]))\Big)\Big\}. \tag{4}$$

Algorithm 1: Proposed CSAC-based Framework

Initialize policy parameters ξ, critic parameters (θ,θ_k) for $\kappa=1,\dots,7$, dual multipliers $\lambda\geq 0$, replay buffer $\mathcal{D}$, and set hyperparameters γ, α, τ, Υ_λ, cost budgets d_k, and maximum episodes.
for episode = 1 to E **do**
  Initialize environment and observe initial state s.
  for each step n in the episode **do**
  1) Sample action $A\sim\pi_\xi(\cdot|s)$.
  2) Execute action A, receive reward $R(s,a)$, costs $\mathbf{C}=[C_1,\dots,C_7]$, and observe next state $s[n+1]$.
  3) Store transition $(s,a,R(s,a),\mathbf{C},s[n+1])$ in $\mathcal{D}$.
  4) Sample mini-batch $\mathcal{B}$ from $\mathcal{D}$.
  5) Update critics $(Q_{R,\theta},Q_{C_k,\theta_k})$ by minimizing errors using Bellman targets and Polyak averaging.
  6) Update policy parameters ξ by maximizing stochastic objective using (5). Set $s[n]\leftarrow s[n+1]$.
  7) **if** iteration number $\tau=0$ **then**
  8) Compute average costs over mini-batch and update dual multipliers using (6).
  9) If $s[n+1]$ is a terminal state, break. **end**
end

Now, we employ eight critic networks, i.e., $Q_{R,\theta}(s,a)$ to estimate the discounted reward and $Q_{C_\kappa,\theta_\kappa}(s,a)$ to approximate the discounted cumulative cost for each constraint. All critics are trained from replay buffer sampled with Bellman targets, and the Polyak averaging is used to stabilize the targets for Lagrangian penalties. At each gradient step, the policy parameters ξ are updated by maximizing the stochastic objective as

$$\mathcal{J}_\pi=\mathbb{E}_{\substack{s\sim\mathcal{B},\\ a\sim\pi_\xi}}\Big\{\alpha\log\pi_\xi(a|s)-Q_{R,\theta}(s,a)+\sum_{\kappa=1}^{7}\lambda_\kappa Q_{C_\kappa,\theta_\kappa}(s,a)\Big\}, \tag{5}$$

where $\mathcal{B}$ is a mini-batch from the replay buffer. The advantage term within the brackets promotes actions that enhance ASR returns while reducing expected costs adjusted by the current multipliers. After every τ actor/critic updates, the multipliers are projected gradient-ascent updated to push the empirical average costs towards their budgets:

$$\lambda_{\kappa+1}=\Big[\lambda_\kappa+\Upsilon_\lambda\Big(\Big(\frac{1}{|\mathcal{B}|}\sum_{(s,a,\mathbf{c})\in\mathcal{B}}c_\kappa\Big)-d_\kappa\Big)\Big]^+, \tag{6}$$

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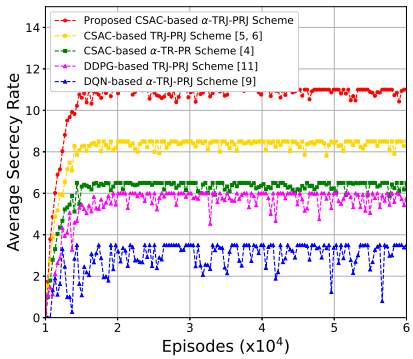

Fig. 2. Convergence behavior of proposed and baseline schemes.

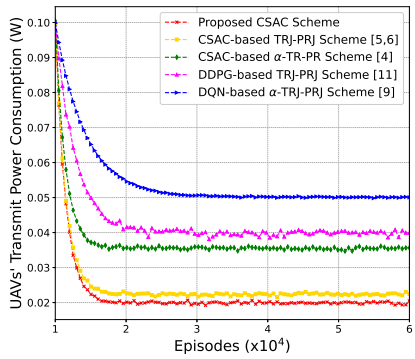

Fig. 3. UAVs' transmit power consumption for different schemes.

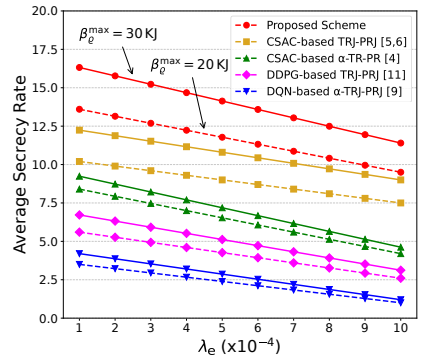

Fig. 4. ASR versus λ_e for proposed and baseline schemes.

with learning-rate Υ_λ. When the cost k exceeds its budget d_κ, the penalty coefficient λ_κ increases, imposing a greater penalty on that cost in future actor updates. Conversely, if the cost is within budget, λ_κ decreases, providing the policy more flexibility. The entropy temperature α can be fixed or adjusted by targeting a minimum expected entropy, preventing early exploitation in the highly non-convex dual-UAV search space. CSAC alternates fast stochastic gradients on (ξ, θ, θ_k) with slower dual updates on λ. Under standard two-time-scale assumptions, these iterations converge to a local saddle point, resulting in an optimal policy.

B. Computational Complexity

During offline training, each gradient update requires one forward-and-backward pass through the actor and eight critic networks. For fully connected layers, a single pass through a network of depth J and widths $\{H_j\}_{j=1}^{J}$ costs $\mathcal{O}(\sum_{j=1}^{J-1} H_j H_{j+1})$. Let C_{NN} denote this term summed over all nine networks, and N_{up} is a total number of stochastic-gradient updates, overall training complexity is $\mathcal{O}(BN_{\text{up}}C_{\text{NN}})$. In online execution, the agent merely samples an action from the actor and evaluates the critics for every τ step to refresh the dual variables. Both operations are forward passes whose cost is linear in network size, i.e., $\mathcal{O}(C_\pi) = \mathcal{O}\big(\sum_{j=1}^{J-1} H_j H_{j+1}\big)$ per time slot. Whereas the total complexity of solving (P1) using the SCA and BCD algorithms, similar to [6]–[8], [10], [11], is given as $\mathcal{O}\big(\big(N^{3.5}\log\big(\frac{1}{\mu}\big) + N^2\nu^{0.5} + (5N)^{3.5}\log\big(\frac{1}{\mu}\big)\big)\psi\big)$, μ is the tolerance level, ψ is the number of iterations, and ν is the number of updates for feasible initial points. The proposed framework is more suited than BCD and SCA approaches regarding computational complexity, scalability, and adaptability to dynamic environments.

IV. Simulation Results and Discussions

We employ a fully connected four-layer neural network architecture for simulations, with each layer consisting of 512 neurons. The model utilizes the Adam optimizer, the ReLU activation function, and the Lagrangian penalty multipliers. Simulation parameters are: $E = 1000$, $\gamma = 0.95$, learning rates for the policy network, critic network, and Lagrangian multipliers are set to 10^{-3}, 10^{-3}, and 10^{-4}, respectively. The UAV propulsion parameters are adopted from [18]. UAVs' initial and final locations are $\mathbf{q}_{U_R 0} = [-500, -400]$, $\mathbf{q}_{U_J 0} = [-500, 400]$ and $\mathbf{q}_{U_R F} = [500, -400]$, $\mathbf{q}_{U_J F} = [500, 400]$, respectively. System parameters are $R_e = 400\,\text{m}$, $\lambda_e = 10^{-4}$, $T = 80$ sec, $\beta_0 = -60$, dB, $N_R = N_D = N_{E_k} = -174$ dB, $P_B = 30$ dBm, $\nu_U^{\max} = 30$, m/s, $H_{\varrho,min} = 50\,\text{m}$ and $H_{\varrho,max} = 200\,\text{m}$, $\beta_\varrho^{\max} = 20\,\text{KJ}$, and $d_{\min} = 50\,\text{m}$, and $P_\varrho = 20$ dBm. The proposed algorithm jointly optimizes $\alpha[n]$, $(q_\varrho[n], H_\varrho[n])$, and $P_\varrho[n]$ using CSAC algorithm (CSAC-based α-TRJ-PRJ scheme). To show the effectiveness of the proposed framework, we consider the following benchmark schemes acquired from the system models of [6]–[8], [10], [11], [13] and utilize the traditional DRL algorithms to solve their formulated problems.

1) CSAC-based α-TR-PR scheme: We optimize $\alpha[n]$, q_R, H_R, P_R without taking U_J in the system [6].
2) CSAC-based TRJ-PRJ scheme: In this scheme, $(q_\varrho[n], H_\varrho[n])$, and $P_\varrho[n]$ are optimized using the SAC algorithm, however $\alpha[m]$ is not considered [7], [8].
3) DDPG-based α-TRJ-PRJ scheme: We optimize $(q_\varrho[n], H_\varrho[n])$, and $P_\varrho[n]$ using DDPG algorithm [13].
4) DQN-based α-TRJ-PRJ scheme: We utilize DQN algorithm for optimizing α, $(q_\varrho[n], H_\varrho[n])$, and $P_\varrho[n]$ [11].

We illustrate the convergence behavior of the proposed CSAC-based α-TRJ-PRJ scheme and its baseline counterparts over training episodes in Fig. 2. It can be observed that the proposed algorithm exhibits the fastest and most stable convergence, achieving a high ASR of around 13.5 bps/Hz. In contrast, the CSAC-based TRJ-PRJ and α-TR-PR schemes converge at the same rate but to lower ASR levels, showing the importance of joint optimization of $\alpha[n]$, $(q_\varrho[n], H_\varrho[n])$, and $P_\varrho[n]$. The DDPG-based and DQN-based approaches demonstrate higher fluctuations and poorer convergence, with the DQN-based scheme exhibiting the worst performance due to its inability to manage the high-dimensional continuous action space effectively. Fig. 3 presents the transmit power consumption trends of the UAVs across training episodes. The proposed CSAC-based α-TRJ-PRJ scheme achieves the lowest transmit power. In contrast, the DDPG and DQN-based schemes consume significantly higher power throughout the learning process, indicating inefficient trajectory and power

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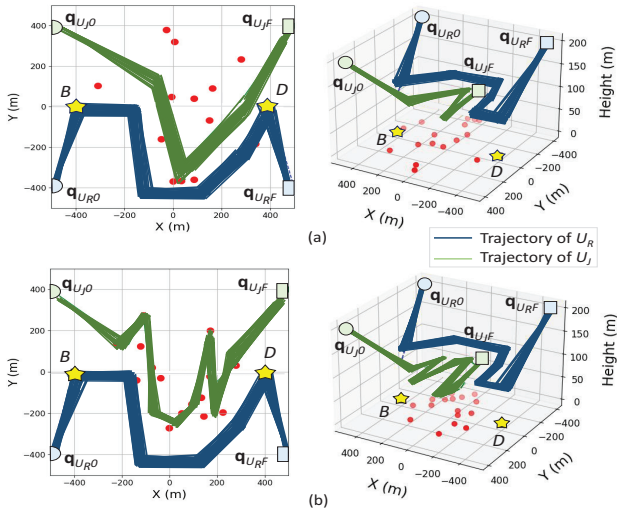

Fig. 5. (a) UAVs' optimized 2D and 3D trajectories for $T = 80$ sec, (b) UAVs' optimized 2D and 3D trajectories for $T = 120$ sec.

control policies. The CSAC-based TRJ-PRJ and α-TR-PR schemes consume more energy than the proposed method due to the non-inclusion of α and U_J in its optimizing framework.

Fig. 4 evaluates the impact of increasing eavesdropper density λ_e on the ASR under different schemes and energy budgets. Two sets of curves are obtained from $\beta_{\varrho}^{\max} = 20$ KJ and $\beta_{\varrho}^{\max} = 30$ KJ, respectively. The ASR value decreases with increasing λ_e for all schemes due to the more serious threat from denser HPPP eavesdroppers. However, the proposed scheme consistently outperforms the baselines across all λ_e values and energy settings. It achieves a steeper ASR improvement with the increased energy budget than the baselines. This is because, at lower $\beta_{\varrho}^{\max}$ values, the power budgets of UAVs are constrained, which prevents U_R and U_J from moving closer to their designated nodes. Conversely, when $\beta_{\varrho}^{\max}$ increases, the additional power budget allows both U_R and U_J to move more quickly toward the ground nodes and eavesdroppers. The optimized 2D and 3D trajectories of the relay and jammer UAVs for different flight duration, i.e., $T = 80$ and $T = 120$ secs, are illustrated in Fig. 5. The proposed scheme enables U_R and U_J to dynamically adapt their trajectories and altitudes to effectively avoid eavesdroppers (red dots) while maintaining strong communication links between S and D by following collision avoidance via (2k). We obtain the optimal UAVs' trajectories and test them 50 times, and these optimal paths are superimposed on a single figure. The UAVs have finer trajectory refinement for $T = 120$ sec, indicating that they utilize extra time to optimize their paths, resulting in more intricate trajectory optimization.

V. Conclusion

This paper addressed the challenges of limited battery life and security in dual UAV-assisted networks, where the data transfer between BS and IoD is accomplished via UAV in the presence of spatially distributed terrestrial eavesdroppers. For this setup, we solved an ASR maximization problem by jointly optimizing the TSR and UAVs' trajectories, altitudes, and transmit powers under practical mobility and power constraints. We formulated the original problem as a CMDP and solved it using a CSAC scheme. Simulation results demonstrated that the proposed scheme achieves superior ASR performance, convergence speed, and UAVs' transmit power consumption compared to the benchmark schemes. Several promising avenues, such as enhancing secrecy in dynamic environments with mobile terrestrial nodes having imperfect CSI or location information, are left for future research.

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